

Unity + FMOD Breathing Integration Handoff

Summary of the investigation, fixes, and remaining issues for the breathing sensor/FMOD integration in TheBreathAwakensVRV2.

Context

- Device input is coming from a microcontroller over serial, with spacebar used to invert the breath sign during development.
- The goal is to map breath sensor input to FMOD parameters controlling a multiband EQ for the breath sound.
- The key FMOD parameters are audioBreathFrequency, audioBreathQVolume, audioBreathGain, and BreathState.

Issues Identified

- Serial port discovery was hardcoded to a single macOS device path, causing most devices to be missed.
- Microcontroller manager OnDisable() crashed when serialized BreathingDeviceData was null.
- BreathFMODDriver was using legacy Input.GetKey() while Unity was configured for the new Input System.
- FMOD breath processing produced constant white noise when there was no input, and the tail cut off too sharply.
- Inhale and exhale behavior was not distinct enough, and intensity mapping was not matching the desired curve.

Fixes Applied

- Updated SerialPortFinder to enumerate available ports, filter likely USB serial devices, and avoid system ttys.
- Guarded MicrocontrollerManager.OnDisable() against null BreathingDeviceData references.
- Found and updated BreathFMODDriver to use sensor data mode by default and to read microcontroller input values.
- Replaced legacy Input.GetKey() with the new Input System API for spacebar detection.
- Added fade logic to avoid hard muting and smooth parameter transitions when breath stops.
- Adjusted parameter mapping ranges to better reflect breath-in and breath-out behavior.

Remaining Problems

- The scene still produces background noise when no breath signal is received, meaning the FMOD event is not fully muted or is still active with low gain.
- Breath intensity scaling still needs tuning so breathe-in and breathe-out volumes follow real input more accurately.
- A reliable path from sensor values to the FMOD EQ curves is required to match the exact 660 Hz → 4.07 kHz and gain/Q behavior.

Next Steps

- Verify the MicrocontrollerManager component has its BreathingDeviceData reference assigned in the Inspector.
- Confirm that the serial port finder only picks actual USB serial devices, and the Try Connect button is opening the port correctly.
- Ensure BreathFMODDriver is using real sensor data mode, not simulated breath-cycle data.

- Tune mapping formulas in BreathFMODEDriver so intensity is driven by the sensor values, not by time-based fade reduction.
- Consider adding explicit idle muting in FMOD when breath input is absent, rather than a background silent event.